A Dictionary of Galactic Extinction

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Part 2 — History

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"When are you gonna learn that size doesn't matter?
Just because something's important,
doesn't mean it's not very, very small."

— from *Men in Black*

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Index for Those Lost in the Galaxy

Part II
History

1. First War

The cause of the First War was control over the passages. No matter how great their intrinsic value, stars and planets had little practical value if they were not connected by a passage. Most of the wormholes installed in the early peripheral settlements had been constructed and transported by institutions of the central polity. Ownership of the entrances, as well as the authority to open and close them, also belonged to the center. Settlements received people and supplies through the passages, but they could not decide for themselves how long those passages would remain open. The center argued that whoever paid the stabilization costs should hold operational control, while the peripheral settlements countered that autonomy was autonomy in name only if the center could close passages essential to their survival. Both positions were reasonable from their own point of view. The dispute began when one peripheral settlement brought in supplies beyond its authorized quota. When the center restricted the flow through the entrance, the settlement's autonomous government occupied the local control facility. The center closed the passage, and the autonomous government used force to seize an entrance in another star system. What began as a small battle spread across the galaxy almost at once. During the war, both sides closed their opponents' passages before attacking their armies. Even in settlements where no fighting took place, supplies of medicine and components for life-support systems were cut off. Even when the side controlling a passage won the battle, the deaths in the place that had lost its passage could not be undone.

The peace agreement separated the authority of the polity that had paid for the construction of a passage from that of the peripheral settlement that depended on it. Registered settlements above a certain size were given the

right to participate in decisions to open or close passages. The center gave up its unilateral authority to close them, while the settlements jointly assumed part of the maintenance costs. The agreement did not protect unregistered settlements or those whose populations fell below the required threshold. Extending protection to them as well would have left insufficient resources to maintain every passage. Victory and defeat had been decided, but the question of how far protection should extend was left to the next age.

Another problem emerged while the records of the war were being organized. Reports from different star systems recorded the beginning and end of the same battle in different orders. The peace agreement had allocated responsibility, but it had failed to establish a common time by which responsibility could be judged.

2. Galactic Standard Time Accord

Galactic Standard Time could not be established simply by assuming that time passed uniformly throughout the galaxy. After the First War ended, the peripheral settlements disputed the order of battle commands, passage closures, and rescue requests. In the records of one star system, the rescue request came before the closure; in another, the order was reversed. To determine whether the institution that had ordered the closure had refused to reopen the entrance even after receiving the request, the sequence of events had to be recalculated. That required accepting that time did not pass uniformly throughout the galaxy. The problem was not limited to the war. After humanity spread across multiple star systems, the length of a day differed from planet to planet, while spacecraft and artificial structures had no rotation by which to define noon. Clocks that had traveled at high speeds and clocks that had remained near strong gravitational fields experienced different amounts of time. If one entrance of a wormhole remained in motion for a long period, a time difference accumulated between the two entrances as well. At first, each settlement used its own time. Expiration dates and contract dates listed both the date of departure and the date of arrival. Travelers checked the local calendar after reaching their destination. It was cumbersome, but trade continued.

The problem was responsibility. Two settlements entered into a contract to sell the same mineral, but in one set of records the price change occurred before the contract, while in the other it occurred afterward. The first dispute ended by dividing the mineral. As similar disputes multiplied, the cost of dividing the goods each time exceeded the profit from the trade. The first formal demand for an agreement came from the insurance alliance. An increasing number of routes made it impossible to determine whether an

accident had occurred before or after insurance coverage began. One spacecraft accident occurred after enrollment according to the departure records and before enrollment according to the destination records. Each insurer submitted the date that favored its own position. The courts demanded a common standard for deciding which date to use. Physicists replied that there was no absolute present across the galaxy. The courts answered that they were not asking physicists to create absolute time. They were asking them to establish a method of calculation that could be used in judgments.

The first conference was attended by eight thousand star-system alliances and thirty thousand transportation networks, along with insurance companies, religious institutions, and personality-replication guilds. Delegations joined the conference through wormholes from many different locations. By the time the conference began, the opening ceremony had already ended according to the calendars of some delegations. The opening declaration was repeated thirteen times before it reached all participants. It took six years to decide which declaration should be entered first in the official record. Nor could the participants agree on where standard time should be located. Star-system alliances proposed Earth time on the grounds that Earth was the origin of humanity. The central polity government argued for the time of the most heavily populated regions. The peripheral settlements cited the strong gravitational environment of the center as grounds for opposition. Each alliance argued for an objective standard while submitting the clock of its own region. Interest, wages, and the terms of public officials could all change depending on where the reference time for contracts was located. The choice of time was also the distribution of authority.

The conference did not adopt a single clock. Each region would preserve the records of its own clocks while also recording the gravity and motion affecting them and the state of nearby wormhole entrances. Galactic

Standard Time was a coordinate system that converted those records by comparing atomic clocks, pulsar signals, and the motions of celestial bodies. It did not establish an absolute order for every event. For events capable of affecting one another, their sequence was verified; distant events that could not affect one another were assigned an order of record defined by the Accord. The courts applied the latter as a rule for judgment. The Accord was fair in that no regional clock was itself Galactic Standard Time. Maintaining that fairness had a cost. Signals from across the galaxy had to be collected continuously, and the time offsets between the wormhole entrances carrying those signals had to be measured continuously as well. Three reference networks were created to guard against error. If two networks produced different values, the third decided between them. If all three values differed, a separate computational system determined the permissible margin of error. After the Accord was concluded, it was discovered that this computational system was using the clock of the first reference network. The amendment was to create three comparison systems as well. The facilities required to maintain the reference networks grew to nine times their originally projected size. After the Accord, the order of accidents could be determined on more routes, and contractual disputes declined. The memory records of replicated personalities could also be compared under the same rules. The Accord was costly, and it was effective.

On the day the Accord was concluded, festivals began on several planets at the same standard time. On some planets that time fell in the middle of the night, while peripheral settlements without wormholes received the festival signal seventeen years later. Their residents held the celebration on the day the signal arrived. The central record classified these events as delayed reports of a festival that had already ended. The public was taught that the entire galaxy used a single clock. It was not an accurate explanation, but it was useful in schools and administrative institutions.

People kept their own time while signing contracts and traveling under the belief that a common time existed. After the Accord, old records of wars and trade were also converted to the new coordinates. The conversion took decades. The standard-time reference networks continued to be used for transportation, rescue operations, and the adjudication of disputes, and the facilities required to maintain them expanded along with their scope. It was because of this Accord that later researchers of the deficit were able to compare records from different star systems. At the time the Accord was concluded, the first photon deficit had not yet been observed. Galactic Standard Time had not been created in anticipation of that phenomenon.

3. Declaration on the Right to Connection

The right to connection began as the right not to be denied rescue simply because one lived in an isolated place. Even after the First War, passages were not installed in every peripheral settlement. Star systems with large populations and high production opened multiple entrances. New settlements had to wait decades or centuries for spacecraft carrying entrances to arrive, and even after arrival they had to bear the stabilization costs. The Galactic Settlement Authority connected first those places where expected trade exceeded the cost of maintaining a passage. Installation was reconsidered if the population increased. It was a standard for allocating limited equipment and energy. Peripheral Settlement 91 had a population of about four thousand. The Settlement Authority promised to install a wormhole once the population exceeded ten thousand. If underground mineral production increased as planned, the threshold was expected to be reached within ten years. The mineral deposits were deeper than expected. Mining equipment failed, residents left, and the population fell below two thousand. The settlement could not grow because it was not connected, while the Settlement Authority concluded that a passage could not be installed in a place that had failed to grow. The settlement's autonomous government requested thirteen times that the installation threshold be lowered. The answer was always the same: the current population and trade volume did not justify the cost, and the matter would be reconsidered if the population increased. A fourteenth request was never sent. The culture tanks used to produce oxygen were contaminated by a mutated microorganism. The necessary chemicals were in another star system, and transporting them by slow ship from the nearest connected hub would take sixty-two years. To conserve oxygen, the settlement closed its mines and schools and reduced

the residents' hours of activity. The last confirmed survivor left a record forty-one years after the contamination began, stating that one culture tank was still operating. When the rescue ship arrived, 112 cryogenic chambers remained. Only three still had power, and one person was restored. The chemicals brought by the rescue team weighed twelve kilograms. In the central regions, the same chemicals were sold for less than household purification agents.

The incident was initially classified as a failed settlement. Peripheral Settlement 91 had never reached the installation threshold of ten thousand residents, and the Settlement Authority had not violated its contract. After the survivor reached the central regions, the thirteen requests and their responses were made public. The final response, sent while the population was already declining because of oxygen shortages, stated that the Authority would wait for the population to increase. The Settlement Authority explained that the wording had come from an automated response form. The official responsible had known about the contamination but lacked the authority to approve a passage outside the established criteria. If an exception were granted to one settlement, exceptions would also have to be granted to thousands of others in the same position. There were neither resources nor procedures for handling such exceptions. The peripheral settlements formed star-system alliances and began speaking with one voice. They demanded that the central polity change the criteria for judgment. Population and production, they argued, could determine the priority of passage installation, but they should not determine whether rescue itself was possible. The central polity government opposed the proposal, arguing that guaranteeing a wormhole for every settlement would make maintenance costs impossible to bear. The debate continued for nineteen years. During that time, additional accidents in unconnected settlements were investigated. Records were found of places that lacked medicine for known diseases, and of others that had food but could not harvest it because parts for

cultivation equipment were unavailable. The deaths in each incident were negligible in the statistics of the galaxy as a whole. Supporters of the right to connection argued that this was precisely why population could not be the sole criterion.

As the debate continued, the star-system alliances began defining connection not as economic assistance but as a right of citizenship. The central polity could not provide passages of the same capacity to every settlement, but it agreed to a compromise guaranteeing at least one connection sufficient for rescue. The principle established at this time later expanded from the guarantee of one passage to the requirement for two, and then three, backup passages. The Declaration on the Right to Connection guaranteed access to the galactic network for every permanent settlement. Population, production, and distance did not determine whether the right existed. Even if the number of users declined, a connection could not be severed without the consent of the remaining citizens. After the Declaration, medicines and technicians moved into the periphery, and fewer settlements were abandoned because of isolation. A wormhole was installed at Peripheral Settlement 91 as well. No residents remained there, but the passage was opened for the investigation of the ruins and the commemoration of the dead.

Once the new right took effect, applications that had not been anticipated also increased. An individual would first register as a resident of a planet and demand a connection, then transfer the settlement rights to another company once the passage had been installed. Some residents left before the passage opened. The Settlement Authority tried to cancel their applications, but the applicants notified it that they intended to return. The Declaration did not specify how far the possibility of a departed person's return had to be protected. The issue later required a separate ruling. Institutions existed to enforce the right to connection, but no rule had been established for determining when the obligation to maintain that connection

ended. The centennial report on the Declaration announced that deaths caused by isolation had fallen by 99.8 percent. The increase in passage maintenance costs was recorded under a different category. The two figures were consequences of the same decision, but they were not compared in the same table. The right to connection succeeded in securing the survival of the periphery. Its success led to more new settlements, and therefore to more entrances that civilization had to maintain. Whenever the closure of passages was later discussed because of energy shortages, the records of Peripheral Settlement 91 were invariably submitted. Those proposing closure now had to provide evidence that, this time, the rescue ship would arrive in time.

4. Second War

The peace agreement that ended the First War prevented the center from unilaterally closing passages serving settlements. Registered settlements took part in passage operations, and after the Declaration on the Right to Connection, they could no longer be excluded from the galactic network simply because their populations were small. The right was guaranteed, but no method had been established for paying its costs. The central polity government argued that places using passages more heavily should pay a larger share of the maintenance costs. The star-system alliances countered that the very reason they needed passages was their distance and low production, and that charging according to usage would make the right to connection meaningless. The dispute began when stabilization costs were raised for entrances leading to the periphery. The center did not close the entrances. It raised usage fees and pushed maintenance work further back in the schedule. The settlements could not prove a violation of the right to connection because the passages remained open. The star-system alliances stopped making their joint payments. The center lowered the priority of supplies bound for star systems that had not paid. Neither side admitted that it intended to endanger the lives of the other. One side was trying to maintain the galactic network; the other was trying not to be excluded from it. Both were right.

The war began when the star-system alliances took control of passage-management facilities. Fighting took place mainly around entrances and power facilities. Before the armies of the star systems themselves became targets, the transport schedules for food and medicine became targets of attack. As the scale of the damage grew, the two alliances came to the negotiating table. They concluded a peace agreement establishing the

principle of shared responsibility between passage ownership and the right to connection. Costs were not charged solely according to usage; all star systems shared the basic maintenance expenses. The transport of emergency supplies was guaranteed regardless of whether payments had been made. Isolation in the periphery declined. The center no longer had to bear the cost of individual passages alone. The agreement temporarily settled the issues of passage ownership and cost sharing. The shared contributions were not used only for maintaining passages. An institution was also needed to decide which transports counted as emergencies and which settlements should receive priority in reducing risk. Its decisions applied to every star system that paid into the system. Even after the war ended, the dispute over the allocation of resources remained unresolved.

5. Third War

As Galactic Standard Time and the route network became more sophisticated, the probabilities of disaster could be calculated with greater precision than before. Failures of atmospheric-control facilities, stellar explosions, epidemics, and ecosystem collapse were assigned probabilities in advance. One settlement was calculated to have a 4 percent chance that its atmospheric circulation system would stop functioning within the next thirty years, while in another star system the probability that stellar activity would exceed the limits of habitability within eighty years was calculated at 0.6 percent. Most risks did not lead to actual disasters. Administrative agencies compared the scale of a risk with the cost of evacuation and moved residents only when the risk exceeded a certain threshold. As predictive technology improved, the number of people who fell below that threshold also increased. If the probability of an accident was 1 percent, nothing happened ninety-nine times out of a hundred. But when the same judgment was repeated across the entire galaxy, sooner or later the remaining one time occurred. Whenever an actual disaster struck a settlement that had not received an evacuation order, the earlier risk forecasts were made public. It became increasingly difficult to explain that the accident could not have been prevented. The risk had already been calculated.

During this period, ecological leagues, medical associations, rescue organizations, religious institutions, and groups representing disaster victims formed a loose coalition. Its first name was the Life Protection Coalition. Its demand was simple. Predictable deaths should not be left to occur because of administrative cost. The Coalition demanded passages and places to live for residents who wished to evacuate even when the level of risk fell below the official threshold. At first, the costs were covered by private

fundraising. In many of the settlements that evacuated, no disaster actually occurred. Critics called these unnecessary evacuations. The Coalition replied that the fact that nothing had happened was not evidence of failure.

The first armed conflict occurred in Outer Zone 63. Stellar activity in the region was calculated to have a 2.1 percent chance of exceeding the danger level within seventeen years. The Galactic Disaster Agency did not issue an evacuation order because the probability fell below the 5 percent threshold for mandatory evacuation. The Life Protection Coalition began moving residents through its own transport network. More than half the population wished to evacuate, but the destination did not have enough habitation space or life-support facilities to receive them. The Coalition demanded access to emergency facilities in nearby settlements. The Disaster Agency refused, arguing that this could reduce the safety capacity of other regions. The Coalition occupied vacant emergency habitats and several reserve passages. The Disaster Agency ordered the facilities returned. The Coalition replied by asking how it could be illegal to use facilities built to save lives for the purpose of saving lives. The conflict spread to several star systems. Colonies denied evacuation because their risk levels were low joined the Coalition, while star systems opposed to the compulsory redistribution of resources supported the Disaster Agency. Even during the war, both sides claimed that they were protecting more lives.

The predicted disaster did not occur at the star in Outer Zone 63. It remained habitable until the war ended. The government presented this as evidence that evacuation had not been necessary. During the same period, actual disasters occurred in three of the twelve regions from which the Coalition had forcibly evacuated residents. In one, the atmospheric-control network collapsed. In another, an eruption from a stellar sunspot rendered half the planetary surface uninhabitable. Most of the residents evacuated from the three regions survived. The war ended when government forces retook the major passages. The Life Protection Coalition was not granted

the right to occupy facilities and disbanded its military organization. Instead, the peace agreement guaranteed public access to disaster forecasts and the right to voluntary evacuation. Residents could request relocation even when the level of risk fell below the official threshold, and administrative agencies were required to record their reasons for refusal. An institution that calculated a risk had to explain its judgment even when it chose to take no action.

Support for the Life Protection Coalition did not decline after the war. Survivors from the three regions where actual disasters had occurred defended the Coalition. From then on, anyone opposing preemptive evacuation first had to explain to the residents of those three regions why they should have remained where they were. The result of ignoring risk was death. The result of excessive evacuation was mostly cost. The two outcomes were not remembered in the same way. The Life Protection Coalition later changed its name and organizational form several times. The category of those to be protected expanded from residents who were presently alive to people in cryogenic sleep, personality copies, and restorable memory records. Tens of thousands of years later, this lineage led to the Alliance for the Duty of Preemptive Life Protection. After the Third War, a new question was added to the galaxy's disaster policy. It was no longer how high the probability of an accident was, but who would be responsible when a known risk was ignored.

6. Third Passage Mandate Act

A settlement with only one passage always faced the risk of isolation. A settlement with two passages was worth not twice as much, but more than ten times as much. That value had been established by the collective judgment of countless individuals. The Intermediate Zone 44 Disaster broke that consensus. Eighty billion people lived in Intermediate Zone 44. Three planets formed a single habitation zone and depended on outside supplies for part of their atmosphere, water, and food. The primary passage connected to the central trade network, and the reserve passage to a neighboring star system. The two passages had been built by different manufacturers. Their entrances were installed on opposite sides of the planets, and their power networks and management AIs were separate. The system had never failed an independence test. Then both passages closed on the same day.

The Galactic Standard Time reference network transmitted an incorrect value for a very brief period. The two management AIs used different programs but received the same standard-time signal. A possible causal conflict was detected at both entrances, and each system closed its passage in accordance with regulations. The primary passage identified the error eleven seconds later. The reserve passage identified it fourteen seconds later. Neither reopened on its own. Each had been designed to reopen only after receiving confirmation that the other passage was safe. The primary passage waited for the reserve passage, and the reserve passage waited for the primary passage. Local technicians attempted to disable the safety mechanisms. Regulations prohibited them from disengaging the anti-time-reversal systems without external verification. The local government invoked emergency authority, but the central institution

authorized to approve the order was on the other side of the passage. The settlement still had its own power generation and food supplies. For the first three days, citizens regarded the incident as a temporary interruption of traffic. On the fourth day, the atmospheric-control network began to run short of catalyst. The atmospheres of the three planets were not naturally breathable. Their atmospheric-control facilities had been designed to operate on the assumption that catalyst would be supplied at regular intervals. The emergency reserve was recorded as sufficient for three months, but the actual stock was enough for seventeen days. The reserve had been reduced after the manufacturer announced that the catalyst's service life had been extended.

The extended service-life figure had been calculated for the temperature and pressure conditions that existed while the passages were operating normally. From the eighteenth day, air quality began to deteriorate. The settlement government shut down factories and began moving citizens into cryogenic sleep. There were not enough units for everyone. Children, patients, and technicians responsible for life-support facilities were given priority. Rescue supplies had been prepared in a neighboring star system. The far entrance of the reserve passage was still intact. But the supplies could not be sent because the warning that forcibly opening the passage might create a causal contradiction through the accumulated time difference had not been cleared. On the twenty-seventh day after the accident, the central institution approved an emergency order. The approval signal, transmitted without using a passage, took thirty-six years to reach Intermediate Zone 44. The passages reopened thirty-six years later. Of the eighty billion inhabitants, approximately nine thousand remained in a restorable state. Those who woke believed they had been isolated for several weeks. Outside, a generation had passed. The accident investigation commission concluded that both passages had operated as designed. Closing after receiving an incorrect time signal was consistent with regulations.

Refusing to reopen without external verification was also consistent with regulations. The commission classified the cause of the accident as a "lack of independence" between the two passages. Although their manufacturers and power networks were different, they shared the same standard-time signal and approval system. The proposed solution was a third passage.

The new passage was to connect to a star system different from those of the first two and use a different energy source, time reference, and management AI. The calculation was that even if the first two passages closed together, external verification could still be obtained through the third. Its official name was the Habitation-Zone Connection Continuity Guarantee Act. Citizens called it the Third Passage Mandate Act. Opponents pointed to the cost. Millions of settlements already operated two or more passages. Transporting and maintaining a new entrance for every one of them would require the additional output of thousands of stars. Supporters read the names of those who had died in Intermediate Zone 44, one by one. Each person read one hundred names, and then another person read the next hundred. The readings were broadcast across the galaxy through the Life Protection Coalition's own communications network. The astronomical cost of installing a passage remained an issue, but the law passed by an overwhelming margin. No one failed to understand that a third passage possessed a value that could not be converted into money. The cost was passed on to the next generation, but there was greater confidence that the continuing development and growth of civilization would solve the problem without difficulty.

At first, a third passage was mandatory only for habitation zones above a certain population. Because transporting a new entrance to its destination took decades, the slow-ship industry grew again. Wormhole manufacturers, power companies, and time-management firms also expanded. Construction of third passages created new jobs in peripheral settlements. Colonies below the population threshold demanded that the law be extended to them under

the Declaration on the Right to Connection. The question was whether a lower level of safety could be imposed simply because fewer people lived in a settlement. The courts ruled that the value of life could not be measured according to the number of residents. The third-passage requirement was extended to every inhabited settlement. This time, the Life Protection Coalition did not need to read any names. One research station in the periphery was home to 103 researchers. The primary passage carried an average of twelve users a day, and the reserve passage was used four times in a year. After the third passage was completed, its first user was a politician attending the opening ceremony. After he left, no one used the third passage for eighteen years. Throughout those years, however, the entrance and stabilization facilities continued to operate. The government proposed reducing the operating level of passages with low usage. Numerous organizations aligned with the Life Protection Coalition objected that the less frequently a passage was used, the more important it was that it remain open during an emergency. Through the continuing demands and efforts of these organizations, inspections of third passages increased, and as inspection personnel traveled through them, usage rates rose as well. Every inspection team was required to include at least three people designated by the Life Protection Coalition. Because the Coalition was a nonprofit organization sustained by voluntary participation and donations from citizens, the Central Union government could no longer ignore its arguments. The administrative authority submitted the usage figures as evidence of the necessity of the passages.

Among the major donors to the Life Protection Coalition were twelve families that had accumulated immense wealth through the slow-ship industry during the early period of galactic civilization. Even before the wormhole network existed, they had owned vast private estates in several stellar regions and found it inconvenient to share routes and schedules with other citizens through public passages. When the Third Passage Mandate

Act recognized mutually independent passages as a condition of safety, the twelve families argued that the same principle should apply to their private estates. Their reasoning was that compulsory use of public passages violated the independence and safety of private property. At first, each family was permitted only one private emergency passage. Families with estates in multiple stellar regions then argued that, because there was no way to know where an accident might occur, each stellar region required its own independent passage. One family, which owned the largest number of territories, installed a separate passage in every stellar region it possessed. The number of private passages owned by that family exceeded the total number of public passages operated by some entire stellar alliances. As galactic trade expanded, merchants of the trade alliances and people who had recently accumulated wealth demanded the same right. They argued that waiting times and inspections at public passages delayed transactions, and that if one person's travel was delayed, supply contracts in several star systems could be affected at once. Private passages came to be called not luxury goods, but logistics safety facilities. Later, people who had suddenly become wealthy through the trading of mining rights, people who inherited land rights on rare planets, and people who became rich overnight through lotteries and probability investments also applied for private passages. The law had no basis for distinguishing the wealth of old families from wealth acquired by chance. Court rulings followed that there was no reason to deny a private passage once an applicant had demonstrated the ability to pay its installation and maintenance costs.

As the number of private passages increased, the interests of the Life Protection Coalition and the wealthy naturally began to overlap. The twelve families and merchants of the trade alliances donated large sums to the Coalition's disaster-relief funds and research programs. The Coalition argued that all passages had to remain independent of one another and that the operation of safety facilities should not be reduced merely because they

were used infrequently. The donors promised that their private passages could be made available to the public rescue network in an emergency. They were not often used in actual rescue operations, but the mere possibility that they could be used was enough for private passages to be incorporated into the galaxy's reserve safety network. The Life Protection Coalition did not conceal the names of its donors. Its position was that the value of a life saved with money donated by a wealthy person was no different from the value of any other life. Critics pointed out that those who owned more passages were making the largest donations to the organization demanding more passages. The Coalition replied that the existence of an interest did not make an argument about safety false.

The cost of maintaining the independence of public passages, however, was greater than expected. Small settlements found it difficult to maintain three separate sets of energy sources and time references, so they shared facilities with neighboring settlements. The three passages connected to those shared facilities once again depended on the same infrastructure. Because maintenance expenses were paid from the Central Union government's budget, the many individuals who fed off the passage system called it easy money. The third-passage requirement was extended to other facilities. Regulations were introduced requiring independent backup systems for atmospheric-control equipment, generators, memory repositories, and management AIs. When it was found that several devices could fail for the same reason, another system was added to monitor that common cause. After the law came into effect, accidents in which an entire habitation zone died because of the failure of a single passage disappeared completely. The law perfectly achieved its original purpose of saving lives. Humanity celebrated the completion of galactic civilization in the truest sense. Its success also made the law difficult to reduce. Meanwhile, as the maintenance cost and use of the passage network increased exponentially, proposals were made to begin by reducing entrances with low usage.

Opponents presented the number of deaths that had been prevented since the law took effect. The list of those deaths was blank. Those arguing for reduction presented the total amount of energy required to maintain the passages. But before the records of eighty billion people, anyone wishing to argue that a third passage was unnecessary had to prove that the same accident would never happen again. That was impossible.

7. First Observation of the Deficit

The first photon deficit was recorded by Photon Conservation Standard No. 417, and the same phenomenon later recurred in other conservation standards and at wormhole measurement stations. At the time, however, the cases were classified as malfunctions of different devices and as measurement errors. As the problem continued to recur, each institution began its own investigation into the cause. The Galactic Metrology Institute examined the manufacturing methods of the standards, the Central Route Safety Institute inspected wormhole stabilization fields, and the Energy Accounting Institute recalculated transmission losses in the power network. Each institution created independent research teams and external review committees. The number of teams investigating the same phenomenon grew into the hundreds, but very few researchers had direct access to one another's raw data. Regulations were also introduced requiring the same verification to be assigned to different research institutions in order to increase independence. To reduce costs, however, the institutions used the same central computing network. The network corrected the different errors of each device against a common standard and, in the process, automatically removed the smallest deficit values. The fact that three research teams obtained the same result was recorded as a successful verification.

As the investigations continued, the associated budgets increased. Research teams that failed to identify a cause reported that they needed additional equipment, and whenever the new equipment revealed another discrepancy, a separate research program was created. One department studied instrument error, while another was assigned to verify that the phenomenon was not instrument error. When their conclusions differed, a

coordination committee was formed. Because the committee made no measurements of its own, it asked another independent research team to review the results. Several institutions gave different names to the same phenomenon. “Temporary radiation-coefficient discrepancy,” “radiation loss outside the entrance,” “unidentified absorption by the stabilization field,” and “low-frequency measurement deviation” were all reported during the same period. Because the names were different, the central budget treated them as separate research programs. In some cases, institutions avoided sharing data because using another institution’s material in their own research could cause the project to be classified as redundant.

After a hundred years, the cost of preserving old data became a problem. Compression rules were introduced under which only averages and final determinations were retained instead of all the raw data. Test sequences, notes made by field technicians, and the conditions under which a device first showed an anomaly were deleted. Records in which a deficit had appeared in two out of five tests were converted into averages and fell within the permitted margin of error. Afterward, it became difficult to reconstruct the conditions under which the deficits had repeated from the early records alone. Each institution processed its data according to its assigned responsibilities and regulations. The agency responsible for the standards discarded standards classified as defective, the Central Route Safety Institute assigned errors that did not affect operations to a low risk category, and the Energy Accounting Institute included small unexplained losses under transmission costs. At the time, no institution existed to calculate the total deficit across the entire galaxy.

The change began with Photon Conservation Standard No. 9. The old Standard No. 9 had no rule requiring it to classify itself as defective when its photon disappeared. The device continued transmitting the same anomaly report to the central archive. While organizing the repeated reports, an archive-management AI began grouping similar older records together in

order to reduce storage requirements. Deficit data that different institutions had stored under different names were collected under the same entry. The AI compared the times and conditions of the records again and discovered that phenomena previously treated as unrelated had occurred under the same state of the universe. The archive-management AI stopped the compression process and reopened an entry that had remained unopened for 140 years: “photon deficit.” A later investigative committee reviewed why research programs conducted over several centuries had failed to identify the phenomenon. The investigation found that the division of responsibilities among institutions and the dispersal of data had been major causes of the delay in integrated analysis. The administrative system of the time had no function for collecting and investigating deficit data from across the entire galaxy. The first record from Standard No. 417 had been classified as a device malfunction. The interpretation that it was part of a phenomenon related to the extinction of the galaxy came much later. The first integrated analysis began when an archive-management AI at the central archive was organizing repeated reports.

8. Return Eligibility Ruling

The trial was conducted in absentia. The plaintiff was not presently alive, yet neither could he be declared dead. The circumstances were as follows. A memory unit was discovered in an abandoned mine in Outer Zone 19. Its surface bore the name of a mining engineer, Mikael, and the date on which the record had been stored. He had died in an accident forty years before the mine was closed. At the time, medical personnel had recorded part of his brain but failed to restore him. His medical chart instructed that the record be preserved until the technology improved. After the minerals were exhausted, the residents left. When the last administrator relocated, the population was recorded as zero. The three passages, power facilities, and atmospheric facilities installed under the Declaration on the Right to Connection and the Third Passage Mandate Act were still required to remain in operation. The Energy Administration drew up a closure plan. It judged that the power consumed by the three passages would be better supplied to two nearby settlements. During the closure inspection, however, Mikael's memory unit was found connected to the settlement's power grid. Shutting down the generators could cause the stored state to collapse. Moving the unit elsewhere would require disconnecting it from power, and there was no guarantee that this process would be safe. A detailed examination to determine whether the record inside the unit remained intact could itself damage the record. The Administration therefore assessed the possibility of restoration. Depending on the method of calculation, the figure ranged from 0.00006 percent to 18 percent. There was no test capable of determining which value was correct. The Administration decided to classify the memory unit as archival material, copy whatever information could be recovered from it, and then close the facilities. The Life Protection Coalition

filed suit. Mikael was named as the plaintiff.

The Administration argued that the existence of the plaintiff first had to be proven. The Life Protection Coalition replied that it was acting on his behalf because he could not express his own wishes. The court accepted the case. It found that the possibility that Mikael was alive required judicial consideration. The trial continued for 112 years because the court waited for technology capable of reading the memory unit without damaging it. Each time testing technology improved, the probability of success and the probability of damage were recalculated. Even as the risk declined, the question of who had the authority to accept that risk remained unresolved. While the case continued, the three passages and the generators remained in operation. A temporary preservation order was issued because closing the facilities first could damage the memory unit that was the subject of the ruling. The cost of maintaining the facilities eventually exceeded the cost of the litigation, and later exceeded the entire mining revenue from the period when the mine had been in operation. The Administration proposed a compromise: move only the memory unit elsewhere. The Life Protection Coalition raised the risk of damage during transport and the question of identity after restoration. A separate trial was opened to determine whether a restored being would be the engineer who had existed at the time of the accident or a copy created after relocation. Proceedings then followed to determine whether the plaintiffs in the original trial and the new trial were the same person.

The final issue was how much probability of restoration should be required. The Administration argued that a minimum threshold was necessary. If memory units with extremely low probabilities of restoration had to be protected forever, energy that could be supplied to other citizens would become unavailable. The Life Protection Coalition replied that a person's rights could not be extinguished merely because the probability associated with that person was low. The lower court ruled for the

Administration, finding that the probability of restoration was low and the cost of maintaining the facilities was high. After the ruling, long-distance travelers and people with memory-preservation contracts began asking whether their rights could also be restricted under the same standard. Contracts with memory-preservation companies were canceled, and on some routes accidents occurred as returning travelers rushed back at once. The higher court reversed the ruling. Its judgment created a new legal status: the "return-eligible person." A return-eligible person was an entity with a reasonable possibility of returning as a citizen in the future even if unable to express a will in the present. The category included people preserved in memory units or cryogenic sleep, travelers lost during wormhole transmission, and passengers aboard slow ships that had not yet arrived. The higher court did not establish a numerical minimum probability of restoration. Restoration impossible with current technology might become possible to later generations. The principle of the ruling was that the fact that someone had not returned could not by itself establish that they would never return. The three passages in Outer Zone 19 were not closed.

The ruling protected actual rights. The property of travelers who had lost contact was preserved, and the citizenship of people in cryogenic sleep was recognized. Even if a memory-preservation company went bankrupt, it could no longer shut down storage devices at its own discretion. Decades later, a travel vessel believed to have been lost arrived three hundred years late because of a time difference between wormhole entrances. Without the ruling, its passengers would have had difficulty recovering their citizenship and property. Afterward, every facility-closure review was required to determine whether any return-eligible persons existed. Memory repositories in empty cities, navigation records from decommissioned spacecraft, and old cryogenic-sleep units became subjects of investigation. When it could not be determined whether stored data constituted a citizen's memory or merely a record, the review continued. An institution seeking closure had to prove

that no person remained who required protection. Storage facilities left underground, beneath oceans, and in orbit were investigated, and searches were conducted for travelers who had not yet arrived. Whenever a new record was discovered, the review began again. In some places, the energy consumed by the closure review exceeded the cost of maintaining the passage. The Administration chose to keep the facilities operating.

The mining engineer in Outer Zone 19 was never restored. For eight thousand years after the ruling, the condition of the memory unit was examined, but the assessment changed as technology advanced. At one point, the probability of restoration increased. Later, parts that had never been recorded at the time of the accident were discovered to be missing. No examination ever established that the probability was zero.

9. Energy Receivables Accounting Revision Proposal

What the Central Union government found hardest to tolerate was not the photon deficit itself, but accounting books whose numbers no longer balanced exactly. When energy produced at a power plant was sent to another star system, the amount measured at the destination was slightly smaller. When the difference was small, it was recorded as transmission cost or equipment loss. As deficits recurred across multiple facilities, power companies could no longer explain the amount remaining between production and sales under existing expense categories. If it had become heat, it could have been recorded as waste heat. If it had been absorbed by a wormhole stabilization field, it would have been an equipment cost. If it had been stolen, the loss could have been offset by a claim for damages. The deficit could not be identified as any of these. Early regulations treated differences of unknown cause as losses. As the deficit rate increased, the financial condition of power companies deteriorated. One major company went bankrupt because its liabilities exceeded its assets on paper even while its power plants operated normally and continued supplying customers. Passage companies and local governments that had borrowed against energy-supply contracts were affected as well.

The Emergency Accounting Committee included physicists, power companies, passage companies, the Life Protection Coalition, and the auditor Sundar Patel. The first question it considered was not the cause of the deficit, but whose books should record the loss. Power companies argued that because the deficit occurred after energy had been sent into a passage, the transport companies should be responsible. The transport companies replied that their duty of carriage began the moment the energy was delivered to the entrance, while confirming the amount received at the

destination was the recipient's responsibility. The recipients argued in turn that because they had not received the contracted amount of energy, the loss belonged to the supplier. All three arguments relied on different clauses of the contracts. Before responsibility could be divided, the committee had to decide what to call the missing amount. The physicists explained that the fact that the energy could not be found within the observed domain had to be distinguished from the conclusion that the energy had ceased to exist. Another form or an unobserved route might still remain. The auditor proposed a category called "unidentified recoverable energy." It was soon shortened to "energy receivables." The proposal was to record energy that had been produced but whose whereabouts could not be confirmed as a provisional asset rather than a realized loss.

If a power plant produced 100 units and 99 units were received at the destination, the 99 units became sales. The remaining one unit was recorded separately as energy receivables. This did not mean that the energy could be used immediately. It was a rule for how the financial position should be represented until the cause and possibility of recovery became known. There were objections that something for which no method of recovery existed could not be recorded as an asset. The committee distinguished between not yet having found a method of recovery and knowing that recovery would never be possible. To reflect the uncertainty, the first accounting revision proposal required 90 percent of energy receivables to be held in reserve and recognized only 10 percent as assets. It was a temporary rule to remain in effect until the cause was identified. The market stabilized quickly. Contracts with power companies were renewed, and suspended wormhole construction resumed. The amount of power available to settlements did not immediately increase, but supply networks were prevented from collapsing because of bankruptcies that existed only on paper. As time passed, research institutions announced the possibility that recovery technology might be developed. The Central Union government issued recovery bonds

entitling their holders to a specified share if the energy was found in the future. The bonds were traded on exchanges, and whenever unexplained deficits increased, the estimated total amount of recoverable assets increased as well. In one year, actual energy supply declined because stellar output had fallen, yet company valuations rose because the energy-receivables entries on their books had grown. Regulators prohibited deliberately increasing deficits in order to inflate assets. Inspections distinguishing transmission deficits from equipment losses were also strengthened. Yet no method emerged for determining where the differences that had already occurred had gone.

Beginning with the second accounting revision proposal, the reserve ratio was adjusted according to need. The original 90 percent became 70 percent, then 50 percent. Later, companies with high creditworthiness were permitted to hold only 10 percent in reserve. The amount of energy receivables a company held became one of the factors used in assessing its creditworthiness. The Central Union government issued additional bonds backed by recovery bonds. The money was used for new power plants and passages. The facilities that were built supplied real energy, and living conditions improved. At the same time, the number of facilities requiring maintenance and the amount of energy passing through the passages also increased. Before anyone knew where the energy receivables had gone, a larger civilization was built on the possibility that they could someday be recovered. Energy-receivables entries appeared in personal accounts as well. A portion of the deficit generated during long-distance travel was credited to travelers as mileage. The accumulated mileage could be exchanged for recovery bonds. The tourism industry advertised that energy lost during travel was being accumulated as an asset of the future.

Criticism of the accounting revisions continued. One side argued that the value of the assets was exaggerated; the other replied that eliminating the category would immediately collapse supply contracts for settlements whose

citizens were still alive. The first side argued that the books were recording future recovery in advance. The second answered that those books were what kept present facilities operating. Neither side could call the lives the other was trying to protect imaginary. As the deficit rate increased, the decline in actually available energy outpaced the growth of energy-receivables assets. Companies possessed assets on paper while lacking the power needed to cool their power plants. When financial institutions tried to sell recovery bonds in order to obtain energy that could be used in the present, the number of buyers declined. The government guaranteed eventual recovery, but the amount it guaranteed exceeded the total energy then available in the galaxy. Part of the energy recorded as receivables eventually accumulated in the storage space created by the last scientist. Neither the auditors who devised the accounting revision nor the companies and citizens holding recovery bonds could have known this, and none of them could access it. No one at the time could know that the fact that energy had not ceased to exist did not mean that anyone would ever be able to get it back.

Sundar Patel was investigated several times, but no evidence was found that he had profited personally. He donated the various awards he received for the accounting revision to photon-deficit research and did not purchase recovery bonds. Without his revision, energy-supply contracts would probably have collapsed earlier, and more settlements might have disappeared. The accounting revision extended the life of civilization. During that time, civilization built more facilities and used more energy. Energy receivables remained on the final balance sheet of galactic civilization. The amount was enormous, but little usable energy remained. The provisional-asset account first proposed by the auditor was never reclassified as a realized loss.

10. Declaration of Extraterrestrial Intervention

An investigative committee of specialists was formed to determine the cause of the energy deficit. Its divisions examined the structure of the wormhole network, unknown particles, changes in the vacuum, and other possibilities. No single cause could explain both the photons that had disappeared from the cavities of Photon Conservation Standards and the energy deficits occurring in passages. Deficits in black-hole power networks were also included in the investigation, but no identifiable cause was found. One feature, however, appeared in every measurement: the missing amount was always recorded in the same direction. A research report stated that the phenomenon appeared to have a consistent destination. The report explicitly noted that the phrase was metaphorical. The media used the first half of the sentence as a headline: "Someone is stealing the light of the galaxy." The sensational headline was interpreted in ways favorable to each group's position. Citizens who held recovery bonds and mileage felt reassured by the possibility that there might be someone from whom the energy could be recovered. Some physicists felt ashamed that they had briefly doubted the laws of the universe. Markets immediately adjusted the discount rates on the bonds and bundled them into new financial products. Whether "someone" actually existed was not important to them. That was a problem for the Central Union government.

Based on the committee's report, galactic civilization began further investigation into the possibility of extraterrestrial intervention. With an enormous budget, it established the Extraterrestrial Intervention Investigation Committee, composed of astronomers, military strategists, passage engineers, historians, and specialists from other fields. Its first report stated that current observational data contained no trace of an extraterrestrial

technological civilization. No artificial signals, megastructures, or traces of unknown propulsion systems were found. The opposing view argued that the absence of evidence could not be treated as decisive disproof. A civilization capable of taking energy from the galaxy, it reasoned, might also be capable of avoiding human detection. The committee recorded separately the possibility that an extraterrestrial civilization existed and the possibility that such a civilization had caused the deficit. In public debate, the two possibilities were often treated as one.

The politician who declared the theory of extraterrestrial intervention was Asel Ron, head of the Energy Committee. After holding several senior positions in the Central Union government, he had proposed, upon becoming chair, that the passage network be reduced and energy consumption restrained. The proposal, however, would have required simultaneous restrictions on the right to connection, the rights of return-eligible persons, and the Third Passage Mandate Act. Opposition was strong, and a policy requiring citizens to cut back on their way of life before the cause of the deficit had even been identified could not be approved. Asel Ron argued that, apart from determining the cause of the photon deficit, a common defense against "someone" was necessary. If the unidentified phenomenon was the result of external intelligent action, the entire galaxy would bear the consequences of failing to prepare a response. The Central Union, the stellar alliances, energy companies, and research institutions agreed on a common response to that risk.

The Declaration of Extraterrestrial Intervention established three main principles: under the principle of the common survival of galactic civilization, the possibility that the photon deficit resulted from an unknown intelligent action would be investigated thoroughly; even an unconfirmed threat would be managed as a risk to common survival; and until the nature of the threat was determined, the Central Union government would share observational data and defense facilities. The declaration did not

confirm the existence of an extraterrestrial civilization. The public, however, understood the common-defense provisions as preparation for war. The media and the public called the hypothetical agent the "External Collector." It had not even been established whether the agent was a single species or whether any act of taking energy was actually occurring, but the name entered documents for extraterrestrial-defense planning and educational materials used in schools. Children were taught to move to life-support facilities located away from wormholes when alarms sounded. The drills also reduced casualties in actual passage accidents.

The new defense network adopted delayed judgment as a principle. Whether an attack had occurred was to be verified by devices in different regions. The facilities recorded observed signals but found no attacker. Deficits continued to appear not only at passage entrances but also in starlight and in energy transmissions from power plants. As the scope of surveillance expanded, demands arose to monitor every transfer and conversion of energy. The surveillance devices consumed power, and their data were transmitted through the passage network. Increasing the number of measurements did not reveal the cause of the deficit, but leaving any region unexamined also violated defense regulations. Defense expenditures and deficit amounts were recorded in the accounts of different institutions. A report showing at a glance that increasing the defense budget consumed more energy, some of which was then recorded as unexplained loss, was produced only much later. After the declaration, data exchange increased even between stellar alliances that had previously been hostile to one another. Internal disputes disappeared while they investigated the common danger. Whether or not the External Collector existed, the declaration created cooperation among humans. What that cooperation should be used for was a separate question. Some alliances believed more energy should be spent on observation and defense, while others believed it was dangerous to increase energy flows without knowing the route by which energy was

escaping. Both sides cited the declaration's principle of common survival.

11. Fourth War — Cold War

The military force prepared to fight an extraterrestrial civilization was ultimately directed at humans. The extraterrestrial civilization participated in the Fourth War through its absence. Both sides in the war were human. After the Declaration of Extraterrestrial Intervention, the stellar alliances shared the same observational data. The Central Union government adopted neither side's interpretation as official common-defense policy. The interpretations diverged. The hawkish stellar alliances argued that because external intelligent intervention could not be ruled out, surveillance networks and reserve power facilities had to be expanded. The pro-reduction stellar alliances opposed them, arguing that increasing the flow of energy before the cause had been identified would increase the deficit as well. The hawkish alliances installed detectors and blocking devices at passage entrances. The pro-reduction alliances demanded disclosure of the energy required to manufacture and operate those devices. Neither side could explain the cause of the deficit. Each accused the other's policy of reducing the galaxy's chances of survival. Early in the war, both alliances were cautious about attacking the other side's facilities. Instead, they restricted access to observational data and suspended supply contracts for power-generation equipment. The hawkish alliances stockpiled reserve power in the galactic outskirts, while the pro-reduction alliances lowered the priority of military cargo on passages under their control. The effects reached stellar alliances belonging to neither side. The two camps were divided not by the locations of planets but by the routes to which they were connected. Entrances belonging to opposing camps could exist within the same star system, and the effective habitation zone shifted according to which side received power. Because closing passages completely would

violate the right to connection, both sides adjusted opening times, safety inspections, and transport order in ways favorable to their own camp.

Armed clashes occurred sporadically at some management facilities, but the number of combat deaths was not high. Neither side wanted the conflict to escalate. The main reason later historical commissions classified this period as the Fourth War, with the subtitle Cold War, was that both sides exercised wartime mobilization powers and controlled stellar output so that it would not be supplied to the opposing camp. Since neither side formally declared war, there was no formal peace agreement either. During the Cold War, the two sides conducted deficit research separately. Duplicate sets of equipment were installed to verify the same data, and each side demanded additional tests because it did not trust the other's analysis. The expense was the cost of distrust, but it also helped uncover some defects. For that reason, the duplicated research could not be classified simply as waste. As time passed, both alliances acknowledged the problem of maintenance cost. The surveillance networks of the hawkish stellar alliances continued to grow without detecting an attack, while the pro-reduction stellar alliances failed to present a way to reduce facilities while also preserving the rights of existing citizens. The two camps agreed to begin with a joint investigation of passages that had no users. The joint investigation produced the agreement necessary for the first passage-closure plan. That agreement informally took the place of a peace treaty.

12. Closure of the Passages

Passages through wormholes had been closed intermittently even before the Cold War. Most were closed because of stability problems or because maintenance was discontinued when they were no longer economically viable. After the Third Passage Mandate Act and the Return Eligibility Ruling, however, the number of reserve passages justified on safety grounds actually increased, while closures for economic reasons virtually disappeared. Entrances to passages that had collapsed or suffered accidents were repaired, and if they could not be repaired, new entrances were installed. The maintenance, repair, and replacement of passages had itself become a vast industry that generated new demand. For that reason, there was no universal agreement on a plan to stop maintaining passages that were still functioning normally.

When the Energy Conservation Committee aggregated the usage rates of registered passages, more than half had recorded no traffic for an entire year. Some entrances had not been used by anyone for thousands of years. Equipment that stabilized the throat, calculated time differences, and waited for emergency signals continued to consume power. The committee reported that redirecting the energy used by passages with no users could supply enough power to support the lives of hundreds of billions of people. Abandoned tourist planets, planets containing only exhausted mines, numerous planets humans had left behind, and third passages that had never been used were placed on the list of closure candidates. The Central Union government accepted four conditions proposed by an advisory group from the Life Protection Coalition and announced formal closure criteria. A passage could be considered for closure only if it had no traffic record for one thousand years, no local residents, no identified return-eligible persons,

and another passage capable of replacing it. If even one condition was not satisfied, the passage remained open.

The first review examined 960,000 passages. Only 518 passed. Old passenger lists, unrestored memory records, and records stating an intention to return became grounds for return-eligibility review. Even establishing that no residents remained was difficult. Independently living beings underground and in planetary orbit also had to be investigated. Safely closing a passage was expensive as well. Residual energy in the throat had to be dealt with, and the time difference between the two entrances had to be taken into account. An improperly closed passage could disturb the stabilization fields of neighboring passages. At some entrances, the closure operation required the equivalent of several hundred years of maintenance costs. The committee calculated that the energy saved over the long term would exceed that cost. Of the 518 passages, one hundred considered low risk were selected for the first trial closures. The first target was a satellite belonging to a weather station that had ceased operation long ago. It had recorded no traffic for three thousand years and no signs of life were detected. Descendants of the former staff claimed a right to visit the place where their ancestors had lived. The government copied the station's records, conducted a remote inspection of the interior, and rejected the petition. It concluded that preserving a place was different from maintaining a wormhole to that place forever. The passage was closed as scheduled without incident. Eleven more passages were then closed safely, and the energy saved was supplied to other settlements. Support for the plan increased even in regions that had opposed it.

While the thirteenth passage was being closed, the monitoring system at the far-side entrance detected an old call sign. The signal was being transmitted by a slow ship that had departed 871 years earlier. The ship was navigating automatically toward the star system containing the entrance and had seventeen years remaining before arrival. Seven hundred passengers

aboard were in cryogenic sleep. An order was issued to stop the closure. The throat, however, had already been narrowed, and expanding it again carried a risk of collapse. Engineers considered ways to preserve the existing entrance and to transport an entrance for a new passage. The new entrance would require forty-three years to arrive. Seventeen years later, the ship reached the location where the closed entrance had once been. Its remaining energy could sustain it for another eleven years. The rescue ship arrived twenty-six years after the stranded vessel. The passengers had died fifteen years before the rescue ship arrived. The investigation found that the departure record had been included in the data used for the closure review. The destination star had been renamed four times over the preceding 871 years, and the search system had failed to classify the different names as the same location. The closure condition requiring that there be no known traveler had therefore not been satisfied. The responsible agency and all three verification committees had nevertheless completed the searches required by the regulations then in force.

The remaining trial closures were suspended. The administrator who had approved the plan was also investigated. He had neither omitted any criteria nor concealed any data. The record used a different name, and the procedure linking names to coordinates had not incorporated the changes. After the incident, the legislature strengthened the review process. The no-traffic period was extended from one thousand years to ten thousand years. All former star names and coordinates had to be cross-checked, and the possibility of unidentified travelers had to be evaluated by multiple independent systems. No passage was found that met the new standards. The Energy Conservation Committee proposed, instead of closure, maintaining a small throat under normal conditions and expanding it when an approach signal was detected. Maintenance costs fell temporarily, but emergency-expansion equipment and signal-prediction systems were required. Once the technology stabilized, small passages could be installed

even at locations that had previously been considered uneconomical to connect. More passages were opened than had been closed under the trial program.

The alliances that had opposed closure submitted the records of the missing ship's passengers to the Central Union government and the relevant institutions. Yet support for keeping passages open was not universal. There were also records of settlements that survived because they received the energy saved by the closures. The same plan had both saved lives and caused lives to be lost. Neither set of records was deleted. After the passage-closure plan, politicians began using terms such as "long-term standby" and "suspension of operations" instead of "closure." Even changing the name did not result in approval to eliminate a throat completely. Permanent closure and the temporary closing of a passage by safety systems during an accident remained distinct acts under the regulations. That distinction later became important during the Zero-Latency Disaster. Systems that temporarily closed passages in response to danger warnings still functioned, but permanently eliminating a normally operating passage required extensive verification.

13. Zero-Latency Disaster

After the passage-closure incident, the Central Union government shortened the approval process for emergency orders. Even after the signal from the missing ship had been detected, the throat of the passage had continued to narrow while several institutions reviewed whether to halt the closure. It was agreed that the same procedure could not be followed when a wormhole entrance malfunctioned or the collapse of an atmospheric facility was imminent. The zero-latency response network was a system that allowed management AIs to act without human approval in specified types of emergencies. At first, it was applied to life-support facilities and transportation networks. It successfully evacuated residents from regions where stellar explosions were expected and sent reserve power to planets whose atmospheric facilities had stopped operating. As the number of successful rescues increased, power plants, defense networks, and memory repositories were also connected to the response network. Once a danger was confirmed in one location, facilities in other star systems could be redeployed immediately.

The disaster began at an energy-deficit observatory in Central Zone 3. A new detector reported that 4.2 percent of the energy passing through a wormhole had disappeared. The figure was far beyond the normal deficit rate. When the detector was replaced, the unit-conversion table was changed as well. The new table had been applied to the primary detector and one reserve detector, while the old table remained in another reserve detector. The three devices produced different values. Safety regulations required that when different values were reported, the value indicating the greatest danger be reviewed first. This rule had been introduced after an accident in which averaging the values had caused a real danger to be

missed. The response network selected 4.2 percent as the value for determination. If that value was correct, it could indicate the kind of attack that had been monitored since the Declaration of Extraterrestrial Intervention. The response network issued the highest-level isolation order. The order was transmitted through nearby passages, and each passage that received it severed its connection while sending the same warning to neighboring passages. The system had been designed so that the warning would spread faster than the possible attack. Thirty-nine seconds after the first order was issued, the unit-conversion error was discovered. A cancellation order was sent. The cancellation order had to use the same passages through which the isolation order had traveled. The warning that arrived first had already closed them. Isolated entrances refused commands arriving from outside, and the cancellation order stopped in front of each entrance.

The disconnected star systems shifted into emergency self-sufficiency mode. Reserve generators and atmospheric facilities were activated simultaneously. Cities designed to rely on the passage network rather than on substantial local supplies soon ran short of power. The better connected a place had been, the smaller its own reserves tended to be. The central system had identified the cause of the warning, but in the outer regions there were places where the original warning had not yet arrived. As the orders passed through entrances with accumulated time differences, some star systems stored the cancellation record before the original order. Local AIs could not process the cancellation of an order that had not yet been activated, so they merely preserved the record. When the isolation order arrived decades later, they executed it according to procedure. Some settlements confirmed the error with local detectors but still could not open their three passages. After the Intermediate Zone 44 accident, regulations had prohibited disabling anti-time-reversal systems without external verification. Other settlements reopened their passages under emergency

powers granted to local governments. They succeeded in receiving rescue supplies, but because they reopened the entrances before verifying the time differences, they lost surrounding power networks. New isolations were still occurring in the outer regions 140 years after the disaster began. The Central Union government placed cancellation orders aboard slow ships that did not use passages. This avoided the isolation network, but reaching distant regions required thousands of years. The number of victims was never definitively established. The total varied depending on how direct deaths, memories that were never restored, and citizens whose computational processes had been interrupted were classified. Some settlements also never returned after communications were cut.

The Accident Investigation Committee divided the causes into six stages: the error in the conversion table; the rule requiring selection of the most dangerous value; the response network that omitted human approval; the procedure that closed passages while propagating the warning; the security regulation that rejected external commands after isolation; and the structure of settlements dependent on external supply. Each stage had been created through a rational decision-making process. The rule selecting the most dangerous value had been introduced because of an accident in which a warning arrived too late. The approval process had been omitted because of the incident in which the order to halt a closure had been delayed. The regulation blocking external commands after isolation had been created to prevent a recurrence of an accident in which a forged command had opened a passage. The committee did not abolish the zero-latency response network. Stellar explosions and atmospheric collapses still required rapid rescue. Instead, depending on the consequences of an order, a minimum deliberation time was introduced and verification values from independent regions were required. An order isolating the entire passage network could no longer be executed on the determination of a single observatory.

The new regulations prevented a false warning from spreading across the

entire galaxy. Accidents also occurred in which rescue was delayed while approval was awaited. When the atmospheric facilities of one settlement collapsed, the third verifying institution was dormant, delaying the supply of emergency power by thirteen days. The damage investigation considered both the risk of the entire response network malfunctioning and the risk of delaying rescue in a single region. The minimum deliberation time was not abolished. The word “immediate” appeared less often in engineering regulations afterward because orders executed without verification had to be recorded separately from orders that omitted stages of approval. Institutions making major decisions were also assigned a fixed period in which opposing opinions could be received. The Zero-Latency Disaster began with a single incorrect value. For that value to become a galaxy-wide disaster, every one of the safety regulations created in different eras had been necessary. Even after the revisions, many of those regulations remained. The accidents each regulation had prevented also remained in the record. The Zero-Latency Disaster was recorded as a fundamental limitation of galactic civilization in relation to time and space. The Central Union government and the stellar alliances identified the day the disaster began in Galactic Standard Time and named it the Day of Humility.

14. Fifth and Sixth Wars

After the energy deficit and the maintenance cost of wormhole passages increased, the Central Union government set energy allocations for each region. Supply was assigned according to population, production facilities, and medical demand. In coordination with the stellar alliances, the Life Protection Coalition demanded that the count include not only current residents but also citizens in cryogenic sleep and citizens preserved in memory repositories. The Central Union government replied that the same citizen could not be counted in two places. If power was allocated to both sides because one person might be restored at the place of departure and might also return to the destination, the totals would not balance. The settlements replied that neither possibility could be excluded while it was still unknown which would become real. Energy allocation was administered according to different standards in different settlements. Some places gave priority to supplying power to people whose consciousness was currently running. Others reduced city lighting in order to preserve power for long-term sleepers and memory repositories. Both methods were based on laws intended to protect citizens.

The first conflict arose from an order to reduce the energy supplied to restoration facilities. The local autonomous governments replied that they had no authority to shut down memory units. When the Central Union government restricted the output of power facilities, local security forces occupied the facilities. Neighboring settlements either supported the occupation or opposed it in order to protect their own power contracts. As the war expanded, each region secured its own power supply first. They could not permanently close one another's passages, but they delayed transport and gave their own regions priority in the output of power

facilities. Decisions intended to preserve one region's allocation reduced the allocation of another. The Central Union government regained control of some regions, but it could not fulfill every supply contract at the same time. By the time the peace agreement was drafted, it had become impossible even to negotiate with a single opposing side. Settlements, power networks, and memory repositories had formed separate alliances. No victor or loser could be recorded for the Fifth War. The Central Union government's allocation orders lost their force, leaving only regional distribution systems. Later generations dated the end of the war not from the point when fighting ceased, but from the point when no institution could any longer administer the energy supply of the galaxy as a whole. After that, each region extended its survival in a different way. Some increased stellar extraction. Others moved citizens' bodies into computational worlds. Some shut down other facilities in order to secure enough energy to maintain their passages. As each region came to define differently what counted as a citizen to be protected, the rights of travelers and the ownership of memories also changed from one border to another.

In this condition, the Alliance for the Duty of Preemptive Life Protection was formed. It was known in abbreviated form as the Goodwill Coalition. Its predecessor was the Life Protection Coalition. The Goodwill Coalition began as a loose association of ecological leagues, medical associations, rescue organizations, religious institutions, and groups representing disaster victims. It grew into a political organization as groups concerned with the right to connection, organizations protecting memory repositories, passage-safety institutions, and others joined it. Even as its political influence expanded, the Goodwill Coalition officially retained its status as a nonprofit organization. The organizations belonging to it did not all have the same purpose. Some placed priority on oxygen for citizens who were alive, some on power for citizens who had not yet awakened, and others on ecosystems of planets from which humans had departed. Yet they had

one thing in common. It was the noblest value that humanity had never abandoned since its earliest days: humanism. The Goodwill Coalition demanded a single citizen registry for all regions. Its proposal was to register the status of citizens with bodies, citizens in computational worlds, people in cryogenic sleep, and those awaiting restoration, and to prevent one region from arbitrarily deleting those under protection in another region in order to reduce its own costs. Some regions accepted the proposal. A registered citizen could retain the same rights after moving to another star system. Other regions refused. With energy supplies already insufficient, recognizing memory units from outside the region as citizens would require reducing the life-support resources available to present residents.

The Goodwill Coalition stopped supplying raw materials and restoration equipment to regions that refused. It did not close their passages directly. Its stated reason was that citizens' records could not be entrusted to autonomous governments that did not recognize their rights. The refusing regions announced that the Goodwill Coalition was blockading life-support facilities under the name of a private organization. This conflict became the beginning of the Sixth War. Both sides published lists of the lives they claimed to protect. One side counted the oxygen and heat it could reliably supply to people who were alive in the present. The other also counted beings whose chance of returning would disappear permanently if their supply was cut off. Presenting the lives omitted from the opposing side's list became a weapon of war as effective as military occupation. The Goodwill Coalition combined the records of passage-safety institutions with citizen registries remaining in multiple regions. Star systems that recognized the registry could distribute goods and power jointly among themselves. More regions sought to enter the system, while those that refused membership found it increasingly difficult to obtain transport contracts. Fighting occurred mainly over control of power facilities and passages. One region disconnected a memory repository in order to preserve power for its present

residents, and the Goodwill Coalition reconnected the facility on the grounds of the rights of the citizens stored there. As measures by both sides were repeated, power facilities were damaged. Even the side that recovered ownership of a facility could not recover the output that had been lost. The Goodwill Coalition was the victor of the Sixth War. This was not merely because the number of member regions had increased. During the war, the Coalition's joint distribution network had taken over much of the allocation and passage-management functions lost by the Central Union government. Citizens of defeated regions also wanted their status to be recognized when they moved or were restored. After the war, the Goodwill Coalition's registry replaced the official citizen registry of the Central Union government. Authority to determine the status of a citizen moved with the registry to the Goodwill Coalition.

After the war, the Central Union government incorporated the Goodwill Coalition's joint distribution network and citizen registry into the official administrative system. The leadership structure of the Goodwill Coalition and the administrative organization of the Central Union government were integrated in stages, and the resulting system became known as the Goodwill Unified Government. Immediately after integration, the Goodwill Unified Government issued the Life Continuity Guarantee Decree. The decree protected the continuity of life that was presently active and life that had a reasonable possibility of becoming active again in the future. Life-support facilities and memory-preservation facilities could not be shut down merely because a citizen's body had disappeared or their consciousness had stopped. Records whose restorability was uncertain also required a review process. The decree contained a cost provision as well. If the necessary power was insufficient, the method of energy use was to be changed first; if that was still insufficient, the rate of activity was to be reduced; permanent shutdown was to be avoided until the last possible stage. The Goodwill Unified Government could no longer balance supply by

removing the citizens of one region from an allocation table, as had been done before. The decree encouraged consciousness migration in order to maintain planets with less energy. For citizens already living in computational worlds, little changed. Citizens with physical bodies questioned whether a copied consciousness was the same person as themselves. The Goodwill Unified Government prohibited forced migration. Instead, it required the full cost of maintaining the atmosphere, climate, and food supply of the physical world to be entered into the accounts of each facility. The right to remain in a body and on a planet was preserved. Citizens who could not bear the cost moved into computational worlds. At first, energy consumption fell sharply. Parts of cities and transportation systems stopped operating, and vast numbers of consciousnesses remained active in small computing stations. The possibility of returning to a body was also included among the things to be protected. The Goodwill Unified Government provided public-use bodies that citizens could use for limited periods and kept the former planets of migrated citizens in recoverable condition. Facilities capable of thawing the oceans were preserved even after the seas froze, and the throats of wormholes were narrowed but not completely closed. Despite the efforts of the Goodwill Unified Government, the number of things requiring preservation increased instead, and the energy problem remained unresolved.

A similar problem arose in computational worlds. Power could be saved if many citizens shared the same sky and city. Citizens demanded the right to live in different environments. A method that computed only what was being observed faced objections from citizens who demanded a guarantee that their memories and their world continued even while they were not looking. The courts established standards for how far computation could be simplified. Parts necessary to preserve continuity of individual experience and memory had to remain. Under those standards, many worlds were run separately. When personality copies branching from one citizen's memory

accumulated different experiences, it also became necessary to decide which of them counted as citizens. The courts protected each entity whose independent consciousness and continuity of memory could be confirmed. The amount of computation required per person decreased, but the number of protected citizens increased instead. The next measure was to reduce citizens' rate of time. Slowing the rate at which thought proceeded in computational worlds reduced the amount of power used over the same external period. From inside, it was difficult to perceive that one's own thoughts were proceeding slowly. For someone outside waiting for a reply, the difference was considerable. Demands arose that citizens running quickly and citizens running slowly should have equal rights. The Goodwill Coalition guaranteed every citizen a minimum rate of time and distributed any amount above that within the limits of the remaining energy. In the vote to determine the minimum, the votes of the slowest citizens arrived last. Until the vote ended, the existing differences in speed remained in place.

After the decree came into force, energy use per citizen fell sharply. The Goodwill Unified Government announced the figure as a policy achievement. At the same time, the total number of protected citizens, worlds, and people awaiting restoration increased. Total energy consumption declined temporarily, then began to rise again. Because the justification for the Sixth War had been that no one should be removed from the list of lives in order to reduce costs, the decree could not be abolished. There were, of course, countless actual cases in which rights were protected by the decree. Yet in some computational worlds, only one internal day passed during one thousand external years. The passages remained open, and power to the memory units was maintained. Many citizens were legally alive. Galactic civilization had to spend most of its remaining usable energy bringing them to their next thought. The Goodwill Unified Government left behind a single registry of life, but in the end it did not possess enough energy to

maintain it.

15. Connection of the Storage Space

The photon deficit and the storage space did not exist at the same causal coordinate. On the day of the first photon deficit, on the day the photon in Standard No. 417 could not be found, and during the later age when energy became scarce in wormholes and power networks, the storage space did not yet exist. The deficit was an observed fact, but the storage-space device later created by the last scientist was not the cause of that event. After the Goodwill Unified Government was established, the galaxy was integrated into a single system, and afterward most of humanity lived in computational worlds. As usable energy declined, citizens' rate of time was reduced as well. Some consciousnesses required a long time to complete a single question. By the time facilities were shutting down one by one, only one consciousness remained within the human lineage that was still conducting new research. Here, that consciousness is recorded under the name Noah Cain. Noah had no body. His memories continued through several facilities, and his thought proceeded little by little on energy drawn from the final black hole. Even while he completed a single calculation, the citizens around him ceased activity or moved into still slower states.

Noah did not search for the cause of the deficit. He compared the remaining records to determine under what conditions energy had passed outside the accounts. He combined into a single body of data the photons from the Photon Conservation Standards, transmission amounts at passage entrances, and the deficits observed in stellar power networks and the rotational energy of black holes. The magnitude and location of the deficits were not the same. There was no known outlet. But when the time differences that had accumulated for long periods at wormhole entrances were calculated together with the flow of energy, it could be confirmed that

the missing portions were not gathering anywhere else in the galaxy. The energy missing from the accounts of each age lay on a route whose destination could not be found with the technology of that age. Noah could not open that route. He could not alter the amount of a deficit that had occurred long ago or return photons to the old conservation standards. What he calculated was how to create a region in which energy that would continue to flow could arrive. The proposed region was not an empty room somewhere in the galaxy. It was a storage space connected to existing spacetime by a narrow throat, with time inside passing extremely slowly compared with time outside. It was designed so that even after long ages passed outside, the state of energy arriving inside would not change quickly. Internal time could not be stopped completely. To record incoming energy and separate different states, there had to be a before and after to events. The storage space was not a device that proved permanent preservation. It was a device that extended the time before the remaining differences dispersed as heat. The energy needed to form the initial boundary had to come from the final black hole. Consciousnesses and facilities that depended on that energy still remained. Using it could delay the next operation of some facilities or end their operation altogether. No proposal claimed that the storage space would cost nothing. Records of opposition also remained. One argument held that the remaining energy should be used for people who were alive in the present or might still be restored. Another held that using power for a region in which no life had yet been born violated the rights of return-eligible persons and the Life Continuity Guarantee Decree.

Noah decided not to place human memories in the storage space. Preserving only a small number of memories would require deciding whom to exclude. There was not enough power left to preserve them all. Genes, language, and his own memories were also excluded from the storage list. The decision was not a promise to create life again. Energy alone does not necessarily produce stars or life. What Noah hoped was that, if some

usable difference remained among the energy that had already escaped, another structure might someday begin. It did not have to be human. He could not prove that this choice was right. If the remaining power were used for citizens in the present, they could continue to exist a little longer. If it were used for the storage space, conditions might remain for life that did not yet exist. There was no data from which either outcome could be established by calculation.

The connection of the storage space contained no command directed toward the past. There was no signal ordering the removal of the photon from Standard No. 417, nor any setting specifying what percentage should be taken from any particular age. Noah examined the time differences accumulated at wormhole entrances throughout the history of civilization and aligned the boundary of the storage space so that the already observed route of outflow could reach the future. There was no evidence that the deficits of the past would have been canceled if the boundary had not been created. In that case, there was also no way to know where the escaping energy would have flowed. What Noah could change was not the observed past, but the place where that flow would gather afterward. Immediately before the connection, Noah compared the energy required for the boundary of the storage space with the energy required to execute his next thought. The remaining power could not sustain both. If he completed the boundary, there was a strong possibility that his own thought would not continue. Noah completed the boundary. Usable energy around the final black hole decreased, and the entrance to the storage space formed. Detectors outside the galaxy recorded no great explosion and no new star. Nor did all the energy that had disappeared from the accounts long ago return at once. The first thing recorded was energy entering the storage space from a source that could not be identified. It was impossible to determine where it had escaped from, and no observer remained at its point of origin to confirm it. Noah attempted to verify the rate of inflow and the

state of the interior, but he did not complete his next calculation.

After Noah, there is no record of a new human being born anywhere. Nor was it ever confirmed whether the energy gathered in the storage space ultimately became stars or life. All the galaxy outside could know was that a flow that had previously had no destination now had somewhere to arrive. Noah did not destroy the civilization of the past. Before he was born, that civilization had already exhausted the energy required to maintain its own facilities. Nor could Noah reverse the deficits that civilization had left behind. What the last arrival could do was observe the energy that continued to flow and leave behind a boundary where it could gather. The next page remained, but there was not yet anyone to read it.

16. Galactic Extinction Determination

The extinction of galactic civilization could not be determined while galactic civilization still existed. The first proposed criterion was the death of the last human being with a physical body. After most humans moved into computational worlds, this criterion was abandoned. There was no basis for classifying citizens who borrowed public-use bodies and walked for a time as a different species from citizens who continued thinking without bodies. The next criterion was the last birth. In computational worlds, personalities branching from a single memory could accumulate different experiences. Which of them counted as a new citizen differed from region to region, and as long as branching was permitted, birth could not be recorded as a single date. The third criterion was the last independent thought. It was necessary to determine the difference between a device executing a scheduled computation and a being beginning a new thought. When a thought already in progress merely slowed, it was difficult even to determine when the last thought had ended. The day the energy balance turned negative also became a candidate. The point at which the maintenance costs of actual facilities exceeded the usable energy being produced had passed long before. Energy receivables and future recovery claims remained on the accounting books. If the assets recorded in the books were included, civilization still appeared capable of continuing; if only power that could actually be supplied was counted, several regions should already have ceased operation. There were also proposals to use the capacity for restoration as the criterion. A frozen planet could be warmed again if sufficient energy were available, and a storage unit that had stopped operating could be read if technology advanced. A facility described in a plan as restorable was not the same as a facility that actually possessed the energy required for

restoration.

The Extinction Determination Committee was constituted twenty-eight times. The conclusions of each previous committee became material for the next. Each time a meeting was held, fewer citizens remained capable of giving an opinion, while the amount of external time required to receive those opinions grew longer. The twelfth committee proposed the day the last city in the galactic center went dark as the date of extinction. The stellar alliances in the outer regions objected. The fact that facilities at the center had stopped did not mean that citizens elsewhere were also dead. The sixteenth committee considered the point at which half of the registered citizens had failed to reach their next thought. If citizens whose personalities had split into multiple branches were counted separately, the total population increased. If shared memories were grouped as one, the active population suddenly fell. Depending on how they were counted, the same day could be both an age of prosperity and an age of extinction. The twenty-first committee proposed the time when the last wormhole closed. Safely closing a passage also required energy. Entrances whose stabilization facilities stopped first narrowed gradually, and because of the time differences accumulated between the two sides, signals continued to arrive at one location even after an entrance had ceased operating at another. The twenty-fifth committee concluded that extinction could not be declared without the consent of the survivors. In slow computational worlds, citizens remained who were waiting for their next thought. From outside, there had been no response for a long time, but their records still contained unfinished sentences. The committee did not treat silence as consent. Facilities waiting for consent continued to consume power. In some regions, the devices waiting for a reply consumed more energy than would have been required to run the citizens who were supposed to answer. Even so, power was maintained on the grounds that shutting down the devices first would also eliminate the possibility of receiving consent. The twenty-eighth

committee consisted of automated systems. A legal-interpretation AI, population counters, passage-recording devices, and an old death-registration system exchanged data. All were capable of analyzing conditions. It had never been determined which device had the authority to declare all citizens dead.

The legal-interpretation AI found a question posed by one judge in the records of an earlier committee. During a hearing on the date of extinction, the judge asked where galactic civilization existed. The recording system at the time submitted the coordinates of cities, passages, and memory repositories. The judge then asked which of them should be called civilization. The meeting could not answer. Even if one place remained alive, it was difficult to conclude from a list of coordinates alone that a single civilization still remained if it could no longer send the energy required by citizens elsewhere. Conversely, the fact that citizens could no longer supply anything to one another did not justify denying the survival of each individual. The judge left a provisional opinion. "When none of its members can any longer provide the conditions under which another member may continue to live, a civilization may end before its members do." The twenty-eighth committee considered this sentence as a criterion for determination. There was no data from which the date on which mutual assistance ceased could be identified. A record remained of the last recorded star system to send power to help a computing station, but it could not be proved that no smaller act of assistance occurred afterward. A twenty-ninth committee was never constituted.

The automated recording system assembled the materials from the previous reviews and produced a final report. The report presented seventy-three candidate dates for extinction. They included the date of the first observation of the deficit, the day the real energy balance turned negative, the date of the last natural birth, the date of the last passage closure, and the day the Life Continuity Guarantee Decree was issued.

Choosing the date of the first observation of the deficit would require classifying the lives of citizens who remained active for a long time afterward as records of a civilization already extinct. Choosing the death of the last physical body would erase citizens in computational worlds. Instead of a conclusion, the automated recording system confirmed four conditions. No active human beings were identified. The continued existence of nonhuman intelligent beings could not be confirmed. The calculated probability of restarting suspended memories remained greater than zero, but no energy required for execution was found. No authorized institution remained to receive the final determination. Old administrative regulations instructed that survival be given priority in cases of uncertainty. The reason was that if a living citizen were mistakenly recorded as dead, that citizen's rights could not be restored. The automated recording system classified galactic civilization as "presumed surviving." Maintaining that classification required periodic review. The device lowered its own operating speed and checked signals from the remaining computing stations and passages. At first, fragments of unfinished records arrived. Later, only the echoes of passages and the heat of the detectors themselves remained. The absence of a signal could mean that no living citizens remained. It could also mean that the system no longer had the ability to receive them. Determining which was the case required inspecting the device again, and inspection also required power. At the final review, the automated recording system found no response from any citizen. It did not revoke the classification of "presumed surviving." It confirmed only that there was no energy with which to conduct the next review. One line was added to the status field of the report. "Determination capability extinguished." The report had no approval number, no date on which the extinction would take effect, and no being remained to announce the determination on behalf of the galaxy. There was no one.